

ibm-granite/**granite-embedding-125m-english**

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Transformers English roberta feature-extraction language granite embeddings mteb

Eval Results (legacy) Eval Results text-embeddings-inference arxiv:2502.20204

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Model card Files xet Community 4

Granite-Embedding-125m-English

News: Granite Embedding R2 models with 8192 context length released.

- [granite-embedding-english-r2](#) (149M parameters): with an output embedding size of 768, replacing granite-embedding-125m-english.
- [granite-embedding-small-english-r2](#) (47M parameters): A first-of-its-kind reduced-size model, with fewer layers and a smaller output embedding size (384), replacing granite-embedding-30m-english.

Model Summary: Granite-Embedding-125m-English is a 125M parameter dense biencoder embedding model from the Granite Embeddings suite that can be used to generate high quality text embeddings. This model produces embedding vectors of size 768. Compared to most other open-source models, this model was only trained using open-source relevance-pair datasets with permissive, enterprise-friendly license, plus IBM collected and generated datasets. While maintaining competitive scores on academic benchmarks such as BEIR, this model also performs well on many enterprise use cases. This model is developed using retrieval oriented pretraining, contrastive finetuning and knowledge distillation.

- Developers:** Granite Embedding Team, IBM

Downloads last month
122,645



Safetensors

Model size 0.1B pa

Tensor type BF16

Inference Provider

Sentence Similarity

Source Sentence

Your sentence here

Sentences to compare

Your sentence here...

Your sentence here...

Add Sentence

Generate

</> View Code

Maximize

Model tree for ibm-granite/granite-embedding-125m-english

Finetunes 6 models

Quantizations 8 models

Spaces using ibm-granite/granite-embedding-125m-english 13

ibm-granite/granite-vision-demo

mteb/leaderboard

maxpar1/leaderboard

mkschulz9/personal-chatbot

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ibm-granite/granite-embedding-125m-english is supported by the following Inference Providers:

HF HF Inference API

View API Code

Dismiss

- **GitHub Repository:** [ibm-granite/granite-embedding-models](#)
- **Website:** [Granite Docs](#)
- **Paper:** [Technical Report](#)
- **Release Date:** December 18th, 2024
- **License:** [Apache 2.0](#)

Supported Languages: English.

Intended use: The model is designed to produce fixed length vector representations for a given text, which can be used for text similarity, retrieval, and search applications.

Usage with Sentence Transformers: The model is compatible with SentenceTransformer library and is very easy to use:

First, install the sentence transformers library

```
pip install sentence_transformers
```

The model can then be used to encode pairs of text and find the similarity between their representations

```
from sentence_transformers import SentenceTransformer

model_path = "ibm-granite/granite-embedding-001"
# Load the Sentence Transformer model
model = SentenceTransformer(model_path)

input_queries = [
    'Who made the song My achy breaky heart',
    'summit define'
]

input_passages = [
    "Achy Breaky Heart is a country song written by Jeffery Atkins and recorded by the band The Judds",
    "Definition of summit for English Language Learners"
]

# encode queries and passages
query_embeddings = model.encode(input_queries)
passage_embeddings = model.encode(input_passages)

# calculate cosine similarity
print(util.cos_sim(query_embeddings, passage_embeddings))
```

[pasupuletkarthiksai/medico-bot](#)

[kiedistv/leaderboard](#)

[malyvena/leaderboard](#)

[HuggingLLM/leaderboard](#)

+ 5 Spaces

Collection including [ibm-granite/...](#)

Granite Embedding  Collection
Embedding mo... • 9 items • U.. •  47

 **Paper for** [ibm-granite/granite-e...](#)

Granite Embedding Models
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 **Evaluation results** 

[mteb/arguana](#) 

ArguAna Default T...   source 58.4*

ArguAna   source 58.4*

accuracy ...	test set	self-reported	67.361
f1 on MTE...	test set	self-reported	55.079
f1_weight...	test set	self-reported	73.551
ap on MTE...	test set	self-reported	17.644
ap_weight...	test set	self-reported	17.644

Expand 7920 evaluations

Usage with Huggingface Transformers: This is a simple example of how to use the Granite-Embedding-125m-English model with the Transformers library and PyTorch.

First, install the required libraries

```
pip install transformers torch
```

The model can then be used to encode pairs of text

```
import torch
from transformers import AutoModel, AutoTokenizer

model_path = "ibm-granite/granite-embedding-125m-english"

# Load the model and tokenizer
model = AutoModel.from_pretrained(model_path)
tokenizer = AutoTokenizer.from_pretrained(model_path)

input_queries = [
    'Who made the song My achy breaky heart',
    'summit define'
]

# tokenize inputs
tokenized_queries = tokenizer(input_queries, return_tensors='pt')

# encode queries
with torch.no_grad():
    # Queries
    model_output = model(**tokenized_queries)
    # Perform pooling. granite-embedding-125m-english uses mean pooling
    query_embeddings = model_output[0][:, 0]

# normalize the embeddings
query_embeddings = torch.nn.functional.normalize(query_embeddings, dim=-1)
```

Evaluation:

The performance of the Granite-Embedding-125M-English model on MTEB Retrieval (i.e., BEIR) and code retrieval (CoIR) benchmarks is reported below.

Model	Paramters (M)	Embedding Dimension	MTEB Retrieval (15)	CoIR (10)
granite-embedding-125m-english	125	768	52.3	50.3

Model	Paramters (M)	Embedding Dimension	MTEB Retrieval (15)	Col (10)
125m-english				

Model Architecture: Granite-Embedding-125m-English is based on an encoder-only RoBERTa like transformer architecture, trained internally at IBM Research.

Model	granite-embedding-30m-english	granite-embedding-125m-english	granite-embedding-107m-multilingual
Embedding size	384	768	384
Number of layers	6	12	6
Number of attention heads	12	12	12
Intermediate size	1536	3072	1536
Activation Function	GeLU	GeLU	GeLU
Vocabulary Size	50265	50265	250002
Max. Sequence Length	512	512	512
# Parameters	30M	125M	107M

Training Data: Overall, the training data consists of four key sources: (1) unsupervised title-body paired data scraped from the web, (2) publicly available paired with permissive, enterprise-friendly license, (3) IBM-internal paired data targetting specific technical domains, and (4) IBM-generated synthetic data. The data is listed below:

Dataset	Num. Pairs
SPECTER citation triplets	684,100
Stack Exchange Duplicate questions (titles)	304,525
Stack Exchange Duplicate questions (bodies)	250,519
Stack Exchange Duplicate questions (titles+bodies)	250,460
Natural Questions (NQ)	100,231
SQuAD2.0	87,599
PAQ (Question, Answer) pairs	64,371,441
Stack Exchange (Title, Answer) pairs	4,067,139
Stack Exchange (Title, Body) pairs	23,978,013
Stack Exchange (Title+Body, Answer) pairs	187,195
S2ORC Citation pairs (Titles)	52,603,982
S2ORC (Title, Abstract)	41,769,185
S2ORC (Citations, abstracts)	52,603,982
WikiAnswers Duplicate question pairs	77,427,422
SearchQA	582,261
HotpotQA	85,000
Fever	109,810
Arxiv	2,358,545
Wikipedia	20,745,403
PubMed	20,000,000
Miracl En Pairs	9,016
DBPedia Title-Body Pairs	4,635,922
Synthetic: Query-Wikipedia Passage	1,879,093
Synthetic: Fact Verification	9,888
IBM Internal Triples	40,290
IBM Internal Title-Body Pairs	1,524,586

Notably, we do not use the popular MS-MARCO retrieval dataset in our training corpus due to its non-commercial license, while other open-source models train on this dataset due to its high quality.

Infrastructure: We train Granite Embedding Models using IBM's computing cluster, Cognitive Compute Cluster, which is outfitted with NVIDIA A100 80gb GPUs. This cluster provides a scalable and efficient infrastructure for training our models over multiple GPUs.

Ethical Considerations and Limitations: The data used to train the base language model was filtered to remove text containing hate, abuse, and profanity. Granite-Embedding-125m-English is trained only for English texts, and has a context length of 512 tokens (longer texts will be truncated to this size).

Resources

- ★ Learn about the latest updates with Granite: <https://www.ibm.com/granite>
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